HAOTIAN LIU

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Education

Xiamen University (XMU) China
Bachelor of Software Engineering, Junior year Sep, 2023 – Jun, 2027

Technical Skills

- ✧ **Programming & Development:** Proficient in Python (automated data processing, model invocation) and Java/JavaEE; experienced in microservices development based on the Spring Boot architecture.
- ✧ **Data & Database:** Skilled in SQL database design and management; strong capabilities in data analysis, feature engineering, and comprehensive model evaluation.
- ✧ **Tools:** Expert in Git, Linux, VS Code; mastered DevOps workflows and Scrum agile development.

Research Experience

MLLM: Role-Playing Consistency Evaluation and Enhancement Sep, 2025 – Present

- ✧ **Benchmark:** Contributed to a fine-grained role-playing benchmark covering 90 core bilingual roles and 100k+ high-quality instruction samples; revealed a 40% performance gap between open-source models (LLaMA, ChatGLM) and GPT-5 in "character-specific knowledge".
- ✧ **Optimization:** Addressed "persona OOC" and knowledge sparsity using Context-Instruct; improved role consistency (CUS) and style mimicry of 7B models by 10.4% via instruction tuning.
- ✧ **Pipeline:** Established a multi-dimensional pipeline (NLP metrics, leaderboards, automated & manual eval); achieved 0.68+ correlation between automated and human assessments with strong OOD generalization.

MLLM: Clinical Reasoning for Brain Disease Detection Oct, 2025 – Present

- ✧ **Reproduction:** Implemented diagnostic reproduction on IntrA (3D point cloud) and TOF-MRA (imaging) datasets, achieving ~70% classification accuracy on test sets.
- ✧ **Reasoning:** Transformed 3D models and imaging features into structured JSON summaries; leveraged VLMs to simulate radiologists' reasoning, identifying 5% occult cases with potential rupture risks.
- ✧ **Efficiency:** Quantitatively validated weak-label schemes, doubling 3D aneurysm labeling efficiency with no significant drop in detection performance.

AI4S: Agent-driven Peptide Functional Prediction Jun, 2025 – Dec, 2025

- ✧ **SOTA Results:** Reproduced the CPPpred-En framework, achieving 97.27% accuracy (MCC 0.945) on the CPP924 dataset, significantly outperforming SiameseCPP.
- ✧ **Agent Pipeline:** Engineered an intelligent Agent pipeline integrating 69 physico-chemical features and 8 PLM embeddings; automated the evaluation of 552 pairs via soft-voting ensemble.
- ✧ **Verification:** Performed full ablation studies and t-SNE visualizations, revealing a 2.19% marginal contribution of physico-chemical features to model accuracy.

Project Experience

Unity AI Interactive Picture Book Sep, 2024 – Feb, 2025

- ✧ **Deployment:** Collaborated with Xiaoqian Tech (National High-Tech Enterprise) from user research to deployment; completed on-site testing at Yanwu Primary School and secured National Software Copyright.
- ✧ **Architecture:** Designed a Multi-Agent system using UnityWebRequest to integrate 3 AIGC APIs (Tongyi, Baidu, etc.) for dynamic generation of text, illustrations, and music.
- ✧ **Engineering:** Led requirement alignment and defined OKRs; established a standardized engineering loop from requirement modeling to version rollback.

Awards

- ✧ Second Prize in XMU Speech Contest ~ Second Prize in Social Practice *4 ~ Class committee member